

Introduction to Machine Learning (ML)

Machine Learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without explicit instructions.

ML algorithms are used in a wide variety of applications, such as in medicine, email filtering, speech recognition, agriculture, and computer vision, where it is difficult or unfeasible to develop conventional algorithms to perform the needed tasks.

Types of machine learning algorithms are traditionally divided into three broad categories: supervised learning, unsupervised learning, and reinforcement learning. Supervised learning algorithms build a mathematical model of a set of data that contains both the inputs and the desired outputs. Unsupervised learning algorithms take a set of data that contains only inputs, and find structure in the data, like grouping or clustering of data points.